Arjuna JEE 2025

Mathematics

Straight Lines

DPP: 4

Q1 Write the equation of the line through the points $(1,-1)$ and $(3,5)$

Q2 Find the measure of the angle between the lines $x+y+7=0$ and $x-y+1=0$.

Q3 Find the equation of the line that has y-intercept 4 and is perpendicular to the line $y=3x-2$.

Q4 Find the equation of the line, which makes intercepts -3 and 2 on the x and y-axis respectively

Q5 Equation of a line is $3x-4y+10=0$ find its slope.

Q6 Find the equation of a straight line parallel to y-axis and passing through the point $(4,-2)$.

Q7 A straight line passes through $(2,3)$ and the portion of the line intercepted between the axes is bisected at this point. The equation of the line is
(A) $2x-3y+12=0$
(B) $2x+3y-12=0$
(C) $3x-2y+12=0$
(D) $3x+2y-12=0$

Q8 The equation of the lines which passes through the point $(3,4)$ and the sum of its intercept on the axes is 14 is:
(A) $4x-3y=24, x-y=7$
(B) $4x+3y=24, x+y=7$
(C) $4x+3y+24=0, x+y+7=0$
(D) $4x-3y+24=0, x-y+7=0$

Q9 If the equation $x\cos\theta+y\sin\theta=p$ is the normal form of the line $\sqrt{3}x+y+2=0$ then values of θ and p are
(A) $\frac{7\pi}{6}, 1$
(B) $\frac{5\pi}{3}, \frac{3}{2}$
(C) $\frac{2\pi}{3}, 1$
(D) $\frac{11\pi}{6}, 4$

Q10 If a line with y-intercept 2 , is perpendicular to the line $3x-2y=6$, then its x-intercept is
(A) 1
(B) 2
(C) 3
(D) 4

Q11 If a perpendicular drawn from the origin on any line makes an angle 60° with x axis. If the line makes a triangle with axes whose area is $54\sqrt{3}$ square units, then its equation is
(A) $x+\sqrt{3}y=18$
(B) $\sqrt{3}x+y+18=0$
(C) $\sqrt{3}x+y=18$
(D) None of these

Q12 The equations of the lines on which the perpendiculars from the origin make 30° angle with x-axis and which form a triangle of area $\frac{50}{\sqrt{3}}$ with axes, are-
(A) $x\pm\sqrt{3}y-10=0$
(B) $\sqrt{3}x+y\pm10=0$
(C) $x+\sqrt{3}y\pm10=0$
(D) None of these

Q13

If the points $(1,3)$ and $(5,1)$ are two opposite vertices of a rectangle and the other two vertices lie on the line $y=2x+c$, then the value of c is-

(A) 4
(B) -4
(C) 2
(D) None of these

Q14 If the coordinates of the points A, B, C be $(-1,5),(0,0)$ and $(2,2)$ respectively and D be the middle point of BC, then the equation of the perpendicular drawn from B to the line AD is

(A) $2x+y=0$
(B) $x+2y=0$
(C) $x-2y=0$
(D) $2x-y=0$

Q15 The lines $x\cos\alpha+y\sin\alpha=p_1$ and $x\cos\beta+y\sin\beta=p_2$ will be perpendicular if

(A) $\alpha=\beta$
(B) $|\alpha-\beta|=\frac{\pi}{2}$
(C) $\alpha=\frac{\pi}{2}$
(D) $\alpha\pm\beta=\frac{\pi}{2}$

Q16 The perpendicular bisector of the line segment joining $P(1,4)$ and $Q(k,3)$ has y-intercept -4. Then a possible value of k is

(A) 1
(B) 2
(C) -2
(D) -4

Q17 If one of the diagonals of a square is along the line $x=2y$ and one of its vertices is $(3,0)$, then its sides through this vertex are given by the equations:

(A) $y-3x+9=0, 3y+x-3=0$
(B) $y+3x+9=0, 3y+x-3=0$
(C) $y-3x+9=0, 3y-x+3=0$
(D) $y-3x+3=0, 3y+x+9=0$

Q18 The equation of a straight line which passes through the point $\left(a\cos^3\theta, a\sin^3\theta\right)$ and perpendicular to $x\sec\theta+y\operatorname{cosec}\theta=a$ is

(A) $\frac{x}{a}+\frac{y}{a}=a\cos\theta$
(B) $x\cos\theta-y\sin\theta=a\cos 2\theta$
(C) $x\cos\theta+y\sin\theta=a\cos^2\theta$
(D) $x\cos\theta+y\sin\theta-a\cos^2\theta=1$

Q19 The equations of the lines through the point $(3,2)$ which makes an angle of 45° with the line $x-2y=3$ are

(A) $3x-y=7$ and $x+3y=9$
(B) $x-3y=7$ and $3x+y=9$
(C) $x-y=3$ and $x+y=2$
(D) $2x+y=7$ and $x-2y=9$

Q20 If m is the slope of a line and $m+2=m+3$, then the possible angle between line and x-axis is

(A) 0°
(B) 90°
(C) 60°
(D) 45°

Q21 Two vertices of a triangle are $(5,-1)$ and $(-2,3)$. If origin is the orthocentre of the triangle, then the third vertex is

(A) $(4,7)$
(B) $(-4,7)$
(C) $(4,-7)$
(D) $(-4,-7)$

Q22 If two vertices of a triangle are $(-2,3)$ and $(5,-1)$, the orthocenter lies at the origin, and the centroid on the line $x+y=7$, then the third vertex lies at

(A) $(7,4)$
(B) $(8,14)$
(C) $(12,21)$
(D) None of these

Q23 Two vertices of a triangle are $(4,-3)$ and $(-2,5)$. If the orthocenter of the triangle is at $(1,2)$, then the third vertex is

(A) $(-33,-26)$
(B) $(33,26)$
(C) $(26,33)$
(D) none of these

Q24

A triangle ABC with vertices $\mathrm{A}(-1,0), \mathrm{B}(-2,3/4)$, and $\mathrm{C}(-3,-7/6)$ has its orthocenter at H. Then, the orthocenter of triangle BCH will be

(A) $(-3,-2)$ (B) $(1,3)$

(C) $(-1,2)$ (D) none of these

Q25 Given vertices $A(1,1), B(4,-2)$ and $C(5,5)$ of a triangle, then the equation of the perpendicular dropped from C to the interior bisector of the angle A is

(A) $y-5=0$ (B) $x-5=0$

(C) $y+5=0$ (D) $x+5=0$

Q26 Let the equations of two sides of a triangle be $3x-2y+6=0$ and $4x+5y-20=0$. If the orthocentre of this triangle is at $(1,1)$, then the equation of its third side is:

(A) $122y-26x-1675=0$

(B) $122y+26x+1675=0$

(C) $26x+61y+1675=0$

(D) $26x-122y-1675=0$

Answer Key

Q1	y-3x+4=0	Q14	(C)
Q2	90°	Q15	(B)
Q3	3y+x-12=0	Q16	(D)
Q4	$-2x+3y=6$	Q17	(A)
Q5	3/4	Q18	(B)
Q6	x=4	Q19	(A)
Q7	(D)	Q20	(B)
Q8	(B)	Q21	(D)
Q9	(A)	Q22	(D)
Q10	(C)	Q23	(B)
Q11	(A)	Q24	(D)
Q12	(B)	Q25	(B)
Q13	(B)	Q26	(D)